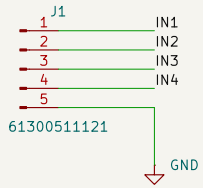
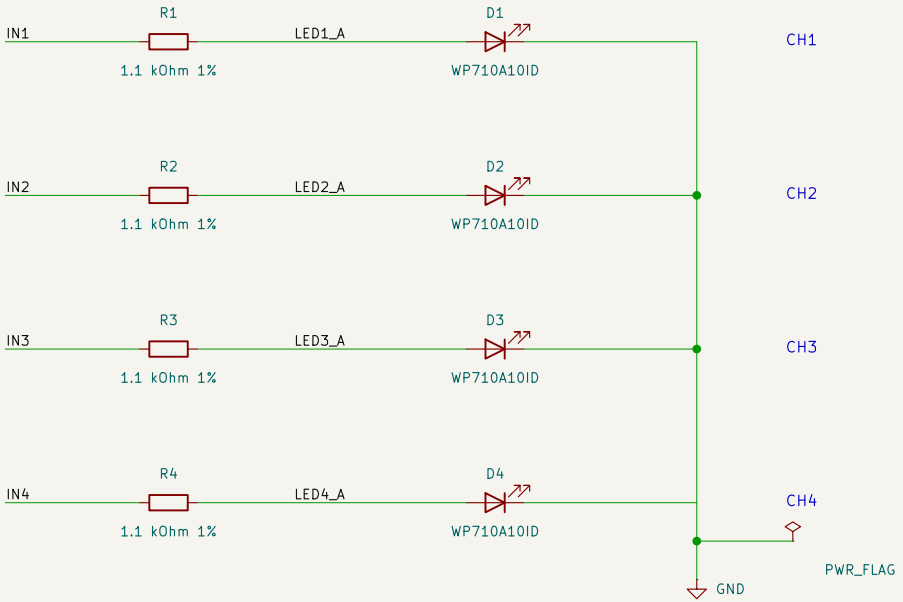
1 INPUTS

Independent 0 to 3.3 V GPIO sources
Common ground; no power pin



2 CURRENT LIMITING + VISIBLE STATUS

Each branch is independent. High input lights its LED.





SOURCES / A1 2026-10-29
Kingbright WP710A10ID pp1-3
Yageo MFR pp2-3
Wuerth 61300511121 p1
Exact bytes: evidence/index.yaml

LED pin1=K; pin2=A.
Header 1..4=IN1..4; 5=GND.

3 DESIGN ASSUMPTIONS / CALCULATED BOUNDS

3.3 V maximum; 1.1 kOhm +/-1%; initial Rmin = 1089 Ohm.
I <= 3.3 / 1089 = 3.031 mA per input; four inputs <= 12.122 mA.
No LED Vf credit is needed for this bound. P(R) <= 10.0 mW.
Room-temperature bench only. Verify actual controller drive limits.
Brightness and fit are unmeasured; no protection or level conversion.

GPIO source limits, physical fit and brightness require review
UNBUILT - NOT APPROVED FOR FABRICATION
PCB Copilot educational example

Sheet: /
File: gpio-indicator.kicad_sch

Title: GPIO INDICATOR | Four independent 3.3 V channels

Size: A4	Date: 2026-10-29	**Rev: A1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1